BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 048 · Flowcharts and Algorithm Design

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. For cost = price × quantity, identify input, processing and output.

Q2. What decides whether an action is legal?
A. Its popularity
B. Its speed
C. Its cost
D. The law

Q3. What is a privacy concern with collecting personal data?

Q4. An online service uses only one language. “Technology cannot affect language or who takes part.” Correct this. Who might be left out?

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I Do · Read new knowledge

Key vocabulary

Flowchart — Diagram of algorithm steps and control flow.

Terminal — Start or End symbol.

Decision — Diamond evaluating a condition.

Process — Rectangle for a calculation or instruction.

Input/output — Parallelogram for accepting/displaying data.

Arrow — Indicates the next stage of flow.

Sub-program — Named procedure represented by a predefined-process symbol.

New learning

Reading a flowchart: A flowchart represents an algorithm visually. Follow arrows from Start through each process or decision until End.

Use the right shapes: Start and End use terminal shapes. Inputs and outputs use parallelograms, processes use rectangles, and decisions use diamonds. Predefined subprograms have their own symbol.

Decisions choose one branch: A diamond evaluates a condition. True/Yes and False/No lead down alternative routes, so a value follows only the appropriate path.

Correct loops and missing paths: Follow the algorithm with a test value. If input is invalid, an arrow can return to input/check for another attempt. Each branch should lead to a defined next action or End.

Worked example

A flowchart checks if a mark is at least 5. If true it prints 'Pass'; otherwise it prints 'Try again'. Explain which branch is followed for 7 and 3. [4 marks]

1. The diamond checks mark >= 5. Only one branch is followed per test.

2. 7 >= 5 is true [1], so the Yes branch prints Pass [1].

3. 3 >= 5 is false [1], so the No branch prints Try again [1].

4. Check both comparisons and the different outputs; avoid running both branches for one input.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

A flowchart checks whether a number is greater than 10. Explain the route and output for 12 and 8. [4 marks]

Worked method: 12 > 10 is true [1], follows Yes and prints High [1]. 8 > 10 is false [1], follows No and prints Low [1].

Guided 2

A lunch-order flowchart repeats a question after an invalid choice. Explain why an arrow loops back to the input. [4 marks]

Worked method: Invalid choice follows the No route [1] and causes another input request [1]. The return arrow rechecks the new choice [1] until it is valid [1].

Guided 3

A flowchart uses a rectangle 'total = a + b' then a parallelogram 'Display total'. Explain both symbols. [4 marks]

Worked method: Rectangle represents processing [1] calculating and storing a sum [1]. Parallelogram represents output [1], displaying the calculated total [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict whether this image file fits.

```
freeMB = 100
imageMB = 80
if imageMB <= freeMB then
    print("Fits")
else
    print("Too large")
endif
```

predict

Question 1

State which flowchart shape represents a decision. [1 mark] Source: Original J277-style teaching question, not an official OCR past paper.

Question 2

A flowchart checks if a customer is at least 16 years old before printing Eligible. Explain what happens for 18 and 14. [4 marks] Source: Original J277-style teaching question, not an official OCR past paper.

Question 3

Describe the purposes of terminal, input/output, process and decision symbols in flowcharts and the use of arrows. [6 marks] Source: Original J277-style teaching question, not an official OCR past paper.

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Exit Ticket · Answer before checking

A flowchart's decision diamond can direct execution down one of two branches.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Inputs: price and quantity. Processing: multiplication. Output: total cost.

Q2. D — The law

Q3. People may lose control over how identifying information is used or shared.

Q4. Technology can change language, access, traditions and who can take part. Consider who benefits or is harmed. Something legal may still be unfair.

I Do / We Do pages · Guided practice

Q1. 12 > 10 is true [1], follows Yes and prints High [1]. 8 > 10 is false [1], follows No and prints Low [1].

Q2. Invalid choice follows the No route [1] and causes another input request [1]. The return arrow rechecks the new choice [1] until it is valid [1].

Q3. Rectangle represents processing [1] calculating and storing a sum [1]. Parallelogram represents output [1], displaying the calculated total [1].

You Do pages · Code First

```
freeMB = 100
imageMB = 80
if imageMB <= freeMB then
    print("Fits")
else
    print("Too large")
endif
```

Expected result: Fits

You Do pages · Independent questions

Q1. A diamond [1].

Q2. 18 >= 16 is true [1] so Eligible prints [1]. 14 >= 16 is false [1], so the not-eligible alternative prints [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

Q3. Terminal identifies start/end [1]; parallelogram input/output [1]; rectangle performs process/calculation [1]; diamond checks condition [1]; it directs to alternative Yes/No branches [1]; arrows show sequence/control flow [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: